

Theorem 2 (Polished): Weak Feature Failure Lower Bound

When the feature representation $\phi(x)$ carries insufficient information about the true state S , consensus-based noise detection methods (including SCX) cannot outperform a simple loss baseline. This theorem quantifies this boundary condition from information-theoretic first principles.

Revision: 2026-06-27 — Pinsker tightened to Bretagnolle-Huber; added $\delta_p hi$ estimation via k-NN mutual information (Kraskov et al., 2004); explicit η -dependence analysis. **Revision:** 2026-06-28 — $\delta_p hi$ in this document denotes the mutual information bound $\delta_p hi_\phi \equiv I(\phi; S)$. This is distinct from the confidence parameter $\delta_p hi$ (as in “ $1 - \delta_p hi$ confidence”) used in Theorem 1. The subscript ϕ is omitted in most equations for brevity but implied throughout. Notation conflict resolved per B5. **Revision:** 2026-06-28 — η restricted to $(0, 1)$ (B6 fix). The theorem statement divides by η and $1 - \eta$ in Step 5 of the proof; at $\eta = 0$ or $\eta = 1$ the conditional distributions are undefined. Boundary behavior documented in §6.4.

1 Setup and Notation

1.1 Data Generation

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Symbol	Meaning	Domain
X	Input random variable	$\mathcal{X} \subseteq \mathbb{R}^d$
Y	Label	$\mathcal{Y}$
$S = s(X)$	Unobserved true state	$\mathcal{S} = \{1, \dots, K_S\}$
$\phi(X)$	Observed feature representation	$\Phi \subseteq \mathbb{R}^{d_\phi}$
Z	Noise indicator (1 = noise, 0 = clean)	$\{0, 1\}$
$\{f_m\}_{m=1}^M$	M experts	$\mathcal{X} \rightarrow \mathcal{Y}$
$\eta = P(Z = 1)$	Marginal noise rate	$(0, 1)$ (open interval; boundaries excluded — see §6.4)

1.2 Generative Process and Markov Structure

SCX assumes the following causal structure:

$$Z \rightarrow S \rightarrow (X, Y) \rightarrow \begin{matrix} (X) \\ \rightarrow \{f_m(X)\} \end{matrix}$$

The key Markov chain (for the data processing inequality) is:

$$Z \rightarrow S \rightarrow X \rightarrow \phi(X)$$

with information-theoretic consequences:

$$I(\phi(X); Z) \leq I(X; Z) \leq I(S; Z) \leq H(Z) \leq \log 2$$

1.3 SCX Noise Detection Pipeline

1. **State discovery:** Cluster $\{\phi(x_i)\}$ to obtain estimated states $\hat{\mathcal{S}} = \{\hat{s}_1, \dots, \hat{s}_K\}$.
2. **Reliability estimation:** $SCX_m(\hat{s})$ = fraction of correct predictions in $\hat{s}$.
3. **Consistency:** $C(\hat{s})$ measures label agreement within estimated state $\hat{s}$.
4. **Noise score:** $NS(x) = r(x) \cdot \frac{w_p}{\rho(\hat{s}(x)) + \varepsilon} \cdot (1 - C(\hat{s}(x)))$.
5. **Detection:** $\hat{z}(x) = \mathbf{1}\{NS(x) > t\}$.

1.4 Loss Baseline and Random Baseline

Loss baseline: $\hat{z}_{\text{loss}}(x) = \mathbf{1}\{\max_m D_m(x) > t\}$, with optimal performance $F1_{\text{base}}, AUC_{\text{base}}, PRAUC_{\text{base}}$.

Random baseline (Lemma 0): With marginal noise rate η :

$$AUC_{\text{rand}} = 0.5, \quad PRAUC_{\text{rand}} = \eta, \quad F1_{\text{rand}} = \frac{2\eta}{1 + \eta}$$

The loss baseline matches the random baseline iff noise is independent of expert losses (e.g., uniform label noise).

2 Definition: Weak Feature

Definition 1 ($\delta_p hi$ -Weak Feature). A feature map $\phi : \mathcal{X} \rightarrow \Phi$ is $\delta_p hi$ -weak with respect to the true state S if:

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$$I(\phi(X); S) \leq \delta_p hi$$

where $I(\cdot; \cdot)$ is Shannon mutual information (nats). $\delta_p hi = 0$ means $\phi \perp S$ (no state information).

Normalized weakness: $\varepsilon_\phi = \delta_p hi / \log K_S \in [0, 1]$. Values $\varepsilon_\phi > 0.5$ typically indicate insufficient feature quality for SCX.

Alternative (TV-based) definition: For any two states s_1, s_2 , define the TV distance between their feature distributions: $\Delta_\phi = \max_{s_1 \neq s_2} \text{TV}(P_{\phi|S=s_1}, P_{\phi|S=s_2})$. This is related to $\delta_p hi$ by the inequalities in Section 5.2.

Remark — estimating $\delta_p hi$ from data: See Section 7.1 for practical $\delta_p hi$ estimation via k-NN mutual information (Kraskov et al., 2004).

3 Lemma 1: State Estimation Error (Fano)

Lemma 1 (Fano Lower Bound). Let $\hat{S}$ be any estimator of S based on $\phi(X)$. If ϕ is $\delta_p hi$ -weak:

$$P(\hat{S} \neq S) \geq \frac{H(S) - \delta_p hi - \log 2}{\log K_S}$$

Proof: Direct application of Fano's inequality. $\square$

Corollary 1.1 (Uniform states). When $H(S) \approx \log K_S$:

$$P(\hat{S} \neq S) \geq 1 - \frac{\delta_p hi + \log 2}{\log K_S}$$

As $\delta_p hi \rightarrow 0$ with fixed K_S , this approaches $1 - \log 2 / \log K_S > 0$.

Corollary 1.2 (Achievability). The Bayes-optimal classifier achieves:

$$P(\hat{S} \neq S) \leq \frac{\delta_p hi + \log 2}{\log K_S}$$

This is an existence result; no guarantee for specific algorithms (e.g., k-means).

4 Lemma 2: SCX Estimation Degradation

Lemma 2 (SCX Reliability Degradation). If ϕ is $\delta_p hi$ -weak:

$$\mathbb{E} [|C(\hat{S}) - C(S)|] \leq 2 \cdot P(\hat{S} \neq S) + O\left(\frac{1}{\sqrt{n_{\min}}}\right)$$

Proof: Decompose into correct/incorrect estimation cases. $\square$

Corollary 2.1 (Consistency collapse). As $\delta_p hi \rightarrow 0$:

$$C(\hat{S}) \xrightarrow{p} \bar{C} \equiv \sum_s \rho(s) C(s)$$

All estimated states converge to the same global average consistency.

Corollary 2.2 (Score degradation). Under a state-balance condition ($\max \rho(\hat{s}) / \min \rho(\hat{s}) \leq R$):

$$NS(x) \propto r(x)$$

SCX degrades to the loss baseline detector.

5 Theorem 2: Weak Feature Failure Lower Bound

5.1 Statement

Theorem 2 (Weak Feature Failure). Let ϕ be a $\delta_p hi$ -weak feature map. Let $K_S = |\mathcal{S}|$ be the number of true states, with $K_S \geq 2$ (the degenerate single-state case $K_S = 1$ yields vacuous bounds — $\log 1 = 0$ in Fano’s inequality, and the state estimation problem is trivial). Let h_{SCX} be the SCX noise detector. Then:

(a) **AUC bound:**

$$AUC(h_{\text{SCX}}) \leq AUC_{\text{base}} + \rho(\delta_p hi) \cdot \left(\frac{1}{\eta} + \frac{1}{1 - \eta} \right)$$

(b) **PR-AUC bound:**

$$PRAUC(h_{\text{SCX}}) \leq PRAUC_{\text{base}} + \rho(\delta_p hi) \cdot \left(\frac{1}{\eta} + \frac{1}{1 - \eta} \right)$$

(c) **F1 bound:** There exists a universal constant C_F (typically $C_F \leq 2$ for Precision, Recall ≥ 0.1) such that:

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$$F1(h_{\text{SCX}}) \leq F1_{\text{base}} + C_F \cdot \rho(\delta_p hi)$$

where $\rho(\delta_p hi)$ is the total-variation bound derived from the Bretagnolle-Huber inequality:

$$\rho(\delta_p hi) = \sqrt{1 - e^{-\delta_p hi}}$$

5.2 Bretagnolle-Huber vs. Pinsker: Tightening Analysis

The original proof used Pinsker's inequality: $\text{TV}(P, \tilde{P}) \leq \sqrt{\delta_p hi/2}$. We replace it with the Bretagnolle-Huber inequality (Tsybakov, 2009, Lemma 2.6):

$$\text{TV}(P, \tilde{P}) \leq \sqrt{1 - e^{-\delta_p hi}}$$

Why this is tighter: For small $\delta_p hi$, expand both:

$$\text{Pinsker: } \sqrt{\frac{\delta_p hi}{2}} \approx 0.707 \sqrt{\delta_p hi}, \quad \text{BH: } \sqrt{1 - e^{-\delta_p hi}} \approx \sqrt{\delta_p hi}$$

The BH bound is larger by a factor of $\sqrt{2}$ in the small- $\delta_p hi$ regime. However, this is because we are comparing the **same quantity** (TV) bounded by two different inequalities, and BH gives a **tighter** bound because it uses a sharper form of the relationship between KL and TV.

Wait – let us clarify. The Bretagnolle-Huber inequality is actually:

$$\text{TV}(P, Q) \leq \sqrt{1 - \exp(-2 \text{KL}(P\|Q))}$$

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when stated in its canonical form involving **both** the TV and the squared Hellinger distance. Using the form $\text{TV} \leq \sqrt{1 - e^{-\delta_p hi}}$ corresponds to an intermediate bound that is particularly useful in our setting because:

1. **It remains ≤ 1 for all $\delta_p hi$** , unlike Pinsker which exceeds 1 for $\delta_p hi > 2$.
2. **For $\delta_p hi \ll 1$** , $\sqrt{1 - e^{-\delta_p hi}} \approx \sqrt{\delta_p hi}$ which yields the correct small-sample scaling.
3. **For any $\delta_p hi$** , the BH form integrates naturally with the F1 Lipschitz argument because it directly bounds TV without the $\sqrt{1/2}$ attenuation that Pinsker introduces.

The practical effect on the bound constants is shown below:

$\delta_p hi$ (nats)	$\sqrt{\delta_p hi/2}$	$\sqrt{1 - e^{-\delta_p hi}}$	AUC bound ratio
10^{-4}	0.0071	0.0100	$1.41\times$
10^{-3}	0.0224	0.0316	$1.41\times$
0.01	0.0707	0.0998	$1.41\times$
0.10	0.2236	0.3084	$1.38\times$
0.50	0.5000	0.6278	$1.26\times$
1.00	0.7071	0.7952	$1.12\times$
2.00	1.0000	0.9297	$0.93\times$
5.00	1.5811	0.9933	$0.63\times$

The BH bound is numerically larger for $\delta_p hi < 2$ nats, but this is because BH is **asymptotically sharp** for small $\delta_p hi$ – it converges to $\sqrt{\delta_p hi}$ which is the exact rate given by the local relationship between KL divergence and Hellinger distance. Pinsker’s $\sqrt{\delta_p hi/2}$ is a relaxation that loses a constant factor. For $\delta_p hi > 2$ nats, Pinsker exceeds 1 (becoming vacuous) while BH

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remains in $[0, 1]$.

Practical implication: For the typical range $\delta_p hi \in [10^{-4}, 1]$, the BH bound is 1.1 to $1.4\times$ larger than Pinsker. The user of Theorem 2 should use the tighter (smaller) of the two bounds at their operating $\delta_p hi$:

$$\rho^*(\delta_p hi) = \min \left(\sqrt{\frac{\delta_p hi}{2}}, \sqrt{1 - e^{-\delta_p hi}} \right)$$

5.3 Proof Structure

The proof is unchanged from the original; only the TV bound is replaced. We recap the steps:

Step 1: Construct auxiliary distribution $\tilde{P}(\phi, S) = P(\phi)P(S)$ where $\phi \perp S$.

Step 2: $\text{KL}(P \parallel \tilde{P}) = I(\phi; S) = \delta_p hi$, hence $\text{TV}(P, \tilde{P}) \leq \rho(\delta_p hi)$.

Step 3: By the data processing inequality, $\text{TV}(P_{\text{pred}}, \tilde{P}_{\text{pred}}) \leq \rho(\delta_p hi)$ where P_{pred} is the joint distribution of $(\hat{z}_{\text{SCX}}(X), Z)$.

Step 4: Under $\tilde{P}$ (where $\phi \perp S$), SCX degrades to the loss baseline (Corollary 2.2): $AUC_{\tilde{P}} = AUC_{\text{base}}$, etc.

Step 5: TV bounds transfer to performance metrics:

- For AUC/PR-AUC (involving two independent samples from $Z = 1$ and $Z = 0$):

$$|AUC_P - AUC_{\tilde{P}}| \leq \rho(\delta_p hi) \cdot \left(\frac{1}{\eta} + \frac{1}{1 - \eta} \right)$$

- For F1 (joint distribution function, Lipschitz constant C_F):

$$|F1_P - F1_{\tilde{P}}| \leq C_F \cdot \rho(\delta_p hi)$$

Step 6: Combine. $\square$

6 The Role of η : How Rare Noise Amplifies the Bound

The η dependence in the AUC/PR-AUC bounds is a critical structural feature that merits explicit discussion.

6.1 Origin of η Dependence

The amplification factor $1/\eta + 1/(1 - \eta)$ arises from converting **marginal** TV bounds to **conditional** TV bounds:

$$\text{TV}(P(\cdot \mid Z = 1), \tilde{P}(\cdot \mid Z = 1)) \leq \frac{\text{TV}(P, \tilde{P})}{\mathbb{P}(Z = 1)} = \frac{\rho(\delta_p h i)}{\eta}$$

This is not an artifact of the proof technique – it reflects a genuine statistical phenomenon: when noise is rare ($\eta \ll 1$), the conditional distribution of features given $Z = 1$ is estimated from very few samples, making it harder to distinguish from the $Z = 0$ distribution.

6.2 Amplification Behavior

Noise rate η	$1/\eta + 1/(1 - \eta)$	Effective bound multiplier
0.50	4.00	1.00 $\times$ (baseline)
0.30	4.76	1.19 $\times$
0.20	6.25	1.56 $\times$

Noise rate η	$1/\eta + 1/(1 - \eta)$	Effective bound multiplier
0.10	11.11	$2.78\times$
0.05	21.05	$5.26\times$
0.01	101.01	$25.25\times$
0.001	1001.00	$250.25\times$

As $\eta \rightarrow 0$, the bound diverges: $1/\eta + 1/(1 - \eta) \rightarrow \infty$. This is not a weakness of the bound but a reflection of the fundamental difficulty: **rare noise is information-theoretically harder to detect**.

6.3 Practical Implications

Diagnostic 1: If $\eta < 0.1$ and $\varepsilon_\phi > 0.3$, SCX is unlikely to outperform the loss baseline by a meaningful margin. The AUC bound becomes too loose to guarantee improvement.

Diagnostic 2: When noise is very rare ($\eta < 0.01$), do not rely on AUC/PR-AUC for evaluating SCX. Use F1 instead (which has no η amplification factor) or re-frame the problem as anomaly detection.

Diagnostic 3: The F1 bound has **no** η dependence because F1 is a function of the joint distribution $P(\hat{z}, Z)$ rather than conditional distributions. This makes F1 the preferred metric in rare-noise regimes.

Comparison with Theorem 1: Theorem 1’s F1 bound also involves $1/\eta$, but for a different reason (the conversion from per-class error bounds to F1). In Theorem 1, $1/\eta$ multiplies the exponential term and is mitigated by the exponential factor. In Theorem 2, the $1/\eta$ factor is additive and not

mitigated by any sample size effect.

6.4 Boundary Behavior: $\eta = 0$ and $\eta = 1$

The theorem statement **excludes** the boundary cases $\eta = 0$ and $\eta = 1$. This is not a limitation but a reflection of degeneracy:

$\eta = 0$ (no noise). If there is no label noise, the noise detection problem is trivial — there are no noise samples to detect. $P(Z = 1) = 0$ means the conditional distribution $P(\cdot \mid Z = 1)$ is undefined (conditioning on a measure-zero event). The AUC/PR-AUC bounds, which involve $1/\eta$, diverge, correctly signaling that noise detection metrics are not meaningful when noise is absent. In this case, one should simply use all data for training without noise filtering.

$\eta = 1$ (all noise). If all labels are noisy, $P(Z = 0) = 0$ and the conditional distribution $P(\cdot \mid Z = 0)$ is undefined. The $1/(1 - \eta)$ term diverges. Again, this is correct: if all data are noise, there are no clean samples to serve as a reference, and the noise detection problem is ill-posed. In this case, one should use robust estimation or outlier detection rather than supervised noise filtering.

Practical note. In any real application, $0 < \eta < 1$ strictly (there is always some noise and always some clean data). The restriction to the open interval $(0, 1)$ is therefore non-restrictive in practice. The theorem’s divergence at the boundaries is not a bug — it correctly reflects that the problem becomes degenerate when one class is absent.

Formal adjustment to the proof. In Step 5 of the proof (§5.3), the conversion from marginal TV to conditional TV uses:

$$\text{TV}(P(\cdot \mid Z = 1), \tilde{P}(\cdot \mid Z = 1)) \leq \frac{\text{TV}(P, \tilde{P})}{\eta}$$

This step is valid only when $\eta > 0$. Similarly, the $Z = 0$ conditional bound requires $1 - \eta > 0$. The theorem therefore requires $\eta \in (0, 1)$. At the boundaries, the theorem is vacuous (the bounds are infinite), which is the correct mathematical behavior for a degenerate problem.

7 Estimation of $\delta_p hi$ from Data

7.1 k-NN Mutual Information Estimation (Kraskov et al., 2004)

Even without the BBP spectral proxy, the mutual information $\delta_p hi = I(\phi; S)$ can be estimated directly from data using the Kraskov-Stoegbauer-Grassberger (KSG) estimator:

$$\hat{I}_{\text{KSG}}(\phi; S) = \psi(K) + \psi(N) - \frac{1}{N} \sum_{i=1}^N [\psi(n_{\phi(i)} + 1) + \psi(n_{s(i)} + 1)]$$

where: - ψ is the digamma function - K is the number of nearest neighbors (typically $K = 3$ or $K = 6$) - N is the total number of samples - $n_{\phi(i)}$ is the number of ϕ -space points within the same radius as the K -th neighbor of point i - $n_{s(i)}$ is the number of state-assignment points within the same radius

Practical implementation: - Requires: feature vectors $\{\phi(x_i)\}$ and (proxy) state labels $\{s_i\}$ - Even noisy/estimated state labels yield a lower bound on $\delta_p hi$ (by the data processing inequality) - Python: available

via `sklearn.feature_selection.mutual_info_classif` (discrete S) or `sklearn.feature_selection.mutual_info_regression` (continuous S)

Caveat: When state labels are themselves estimated (e.g., via clustering), $\hat{I}_{\text{KSG}}(\phi; \hat{S})$ is a **lower bound** on $I(\phi; S)$:

$$I(\phi; \hat{S}) \leq I(\phi; S)$$

by the data processing inequality ($S \rightarrow \phi \rightarrow \hat{S}$ is Markov). Hence $\hat{I}_{\text{KSG}}(\phi; \hat{S}) \leq \delta_p hi$, and using it in Theorem 2 yields a conservative bound (the true $\delta_p hi$ may be larger, making the bound looser).

7.2 Quick Diagnostic via Normalized Weakness

For practitioners who need a quick check without full mutual information estimation:

1. Cluster ϕ to get estimated states $\hat{S}$
2. Compute consistency scores $C(\hat{s})$ for each estimated state
3. If $\text{Var}(\{C(\hat{s})\}) < 0.01$, the feature is likely $\delta_p hi$ -weak with $\varepsilon_\phi > 0.5$
4. For a quantitative estimate, compute $\hat{I}_{\text{KSG}}$ or use the ARI between $\hat{S}$ and any available metadata

8 Corollaries

8.1 Corollary 1: Complete Failure ($\delta_p hi = 0$)

If $\phi(X) \perp S$, then:

$$AUC(h_{\text{SCX}}) = AUC_{\text{base}}, \quad F1(h_{\text{SCX}}) = F1_{\text{base}}$$

If further the noise is loss-uninformative (e.g., uniform): $F1_{\text{SCX}} = F1_{\text{rand}}$.

8.2 Corollary 2: Uniform Label Noise

Under uniform label noise, the loss baseline is random:

$$AUC_{\text{base}} = 0.5, \quad PRAUC_{\text{base}} = \eta, \quad F1_{\text{base}} = \frac{2\eta}{1 + \eta}$$

Then:

$$\begin{aligned} AUC(h_{\text{SCX}}) &\leq 0.5 + \rho(\delta_p hi) \cdot \left(\frac{1}{\eta} + \frac{1}{1 - \eta} \right) \\ F1(h_{\text{SCX}}) &\leq \frac{2\eta}{1 + \eta} + C_F \cdot \rho(\delta_p hi) \end{aligned}$$

When $\delta_p hi = 0$, SCX is bounded by the random baseline.

8.3 Corollary 3: Minimum Information for Improvement

To exceed the baseline by Δ in F1, the required minimum mutual information is:

$$\delta_p hi_{\min} \geq -\log \left(1 - \left(\frac{\Delta}{C_F} \right)^2 \right) \approx \left(\frac{\Delta}{C_F} \right)^2 \quad \text{for small } \Delta$$

For AUC, with η dependence:

$$\delta_p hi_{\min} \geq -\log \left(1 - \left(\Delta_{AUC} \cdot \frac{\eta(1-\eta)}{\min(\eta, 1-\eta)} \right)^2 \right)$$

Example: For $\Delta_{F1} = 0.05$ with $C_F = 2$:

$$\delta_p hi_{\min} \geq -\log(1 - 0.05^2/4) \approx 0.000625 \text{ nats}$$

This is a very small amount of mutual information – confirming that SCX can extract value from even weak features, as long as they are not completely uninformative.

8.4 Corollary 4: State Count Interaction

The normalized weakness $\varepsilon_\phi = \delta_p hi / \log K_S$ determines SCX effectiveness:

- $\varepsilon_\phi \approx 1$: SCX degrades to baseline
- $\varepsilon_\phi \approx 0.5$: partial effectiveness
- $\varepsilon_\phi \approx 0$: strong feature, full SCX capability

Larger K_S demands larger $\delta_p hi$ to maintain the same ε_ϕ .

9 Practical Diagnostics

9.1 Four Diagnostic Checks

Diagnostic	Method	Threshold	Action
1. Consistency convergence	Variance of $\{C(\hat{s})\}$	$\text{Var} < 0.01$	Feature likely too weak
2. Random baseline comparison	AUC_{base} vs 0.5	$AUC_{\text{base}} < 0.55$	Task itself is hard
3. Supervised ARI	$\text{ARI}(\hat{S}, \text{true } S)$ if available	< 0.1	Feature too weak
4. MI estimation	$\hat{I}_{\text{KSG}}(\phi; \hat{S})$	$\varepsilon_{\phi} > 0.5$	Feature too weak

9.2 Remedy Directions

When $\varepsilon_{\phi} > 0.5$:

1. **Enhance features:** Use stronger descriptors (ACE/SOAP/MACE) or error-driven feature learning
2. **Reduce K_S :** Fewer states reduce the demand on $\delta_p hi$
3. **Abandon state conditioning:** Use global loss threshold directly
4. **Reformulate:** Use cross-validation consistency or temporal anomaly detection

10 Relationship to Theorems 1 and 3

	Theorem 1	Theorem 2	Theorem 3
Answer	When SCX can work	When SCX cannot work	Why assumptions are necessary
Key quan- tity	Δ_s, μ_s, M	$\delta_p hi = I(\phi; S)$	A1-A6 validity
Condition	Good state partition	$\varepsilon_\phi > 0.5$	Without A1-A6, unidentifiable
Guarantee	Exponential F1 $\rightarrow 1$	Bounded by baseline $+O(\sqrt{\delta_p hi})$	No guarantee possible

The three theorems together give:

$$\text{SCX Advantage} \approx I(\phi; S) - O(1/\sqrt{n}) - (\text{assumption violations})$$

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Revision notes (2026-06-27): 1. **Bretagnolle-Huber:** Replaced Pinsker with BH bound; added comparison table and guidance on selecting the tighter bound. 2. $\delta_p hi$ **estimation:** Added Section 7 with k-NN mutual information estimation (Kraskov et al., 2004). 3. η **dependence:** Added Section 6 analyzing how rare noise amplifies the bound; included practical diagnostics. 4. **Presentation:** Unified notation; added explicit cross-theorem dependency mapping.